



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

1. COURSE: DATA ANALYTICS AND VISUALIZATION - 24DSB3101

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TITLE OF THE PROJECT	Intelligent Customer Churn Prediction Using Data Analytics and Classification Models	
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Abstract

Customer churn is a major challenge for organizations, as losing existing customers can negatively affect customer relationships and business performance. Machine learning models can be used to predict whether a customer is likely to churn by analyzing demographic, service, usage, contract, and billing-related information. However, customer datasets often contain numerous variables and complex patterns, making it difficult to identify the key factors associated with churn through traditional analysis alone. To address this challenge, this project proposes a data analytics and machine learning framework for intelligent customer churn prediction by integrating Exploratory Data Analysis (EDA), preprocessing, feature engineering, classification models, risk scoring, and visualization.

Initially, the customer churn dataset undergoes data preprocessing, including missing-value handling, categorical data encoding, normalization, and class balancing using SMOTE. A detailed Exploratory Data Analysis (EDA) is then performed to understand churn distribution, customer patterns, correlations, outliers, and important relationships within the data. Feature engineering and feature selection are applied to identify relevant customer, service, usage, contract, and billing factors associated with churn.

Following preprocessing and EDA, five machine learning classification algorithms, namely Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and evaluated. The models are compared using standard performance metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The trained models generate churn probabilities, which are further categorized into Low, Medium, and High churn-risk levels to support customer prioritization.

To improve the interpretability of the predictions, the framework incorporates SHAP (SHapley Additive exPlanations) to identify the most influential factors contributing to customer churn. SHAP provides both global feature importance and local explanations for individual customer predictions, helping stakeholders understand why a customer is classified as likely to churn. The analytical results are presented through visualizations and an interactive dashboard containing model performance, important churn factors, customer risk categories, and prediction results.

Finally, a comparative analysis is conducted across all five machine learning models to evaluate their performance and usefulness for customer churn prediction. The proposed framework combines data

analytics, machine learning, explainable AI, risk scoring, and visualization into a unified customer churn analysis pipeline. The project aims to provide accurate and interpretable churn predictions that can support data-driven customer retention and decision-making.

Signature of the Guide

HOD